Can skill and practice, mixed, produce each dataset's change?

16 contrasts in 11 datasets, trials pooled over units; the extended Wong-Wang model calibrated to A, then changed to fit B.

Question	Measured	Answer
6. Skill + practice produce the observed change adequately (p > .05)	6 of 16 contrasts	a few
7. Where they do, what is needed	both: 3; either: 1; practice: 1; skill: 1	per contrast (Figure 7)
8. Where they do not, the single addition that makes the fit adequate (lowest AIC)	drive: 7; ndt: 2; none adequate: 1	drive
adequate with each addition (of the failures)	drive: 8 of 10; ndt: 4 of 10; retrieval: 0 of 10	
adequate with drive level and non-decision time both added (of the failures)	10 of 10	
9. Speed-accuracy benchmarks need more than skill + practice	2 of 2	both

Per contrast

Id	Task	Change	Instance share in A	Skill + practice	Lowest AIC
P1†	brightness discrimination	Time on task	0%	no; + drive	SP + drive
P2	lexical decision	Time on task	8%	practice	practice
P3	lexical decision (BLP)	Time on task	33%	both	SP + drive
P4	motion discrimination	Time on task	0%	no; + drive	SP + drive
P5	visual search	Time on task	42%	either	practice
P6	visual working memory	Time on task	6%	both	skill + practice
P7	Stroop and Simon	Time on task	0%	no; + ndt	SP + ndt
P8	cued task switching	Time on task	3%	both	SP + drive
P9	attention network (flanker)	Time on task	0%	skill	SP + drive
A1†	brightness discrimination	Instruction	0%	no; + drive	SP + drive
A2	lexical decision	Instruction	0%	no; + drive	SP + drive
F1	lexical decision	Word frequency	10%	no; + drive	SP + drive + ndt
S1	StarCraft 2 by league	Expertise	58%	no; none adequate	SP + drive + ndt
S2	motion discrimination	Expertise	0%	no; + ndt	SP + ndt
D1†	motion discrimination…	Stimulus strength	33%	no; + drive	SP + drive
D2	motion discrimination	Stimulus strength	5%	no; + drive	SP + drive

1. Skill = drive spread u0_sd changed with mean drive fixed; practice = stored instances logN changed; drive level = u0 and u0_sd scaled together; non-decision time = ter_mu shifted; retrieval speed = retrieval onset scale s_W.

2. Adequate: chi-square over the 5th to 99th percentiles' percent change, with data (bootstrap over units) and simulation error, p > .05. AIC = chi-square + 2 x changed parameters.

3. Instance share: % of A's simulated trials won by the retrieval route; practice can act only through these trials.

4. † Two or three units (Ratcliff, three participants; Roitman, two monkeys): the bootstrap over units gives wide intervals, so the adequacy test is weak there.

5. Sources: 4_output/mixture_summary.csv, mixture_fits.csv, pooled_profiles.csv. Model: WW-RTShapeAnatomy/model/wwsim.py.

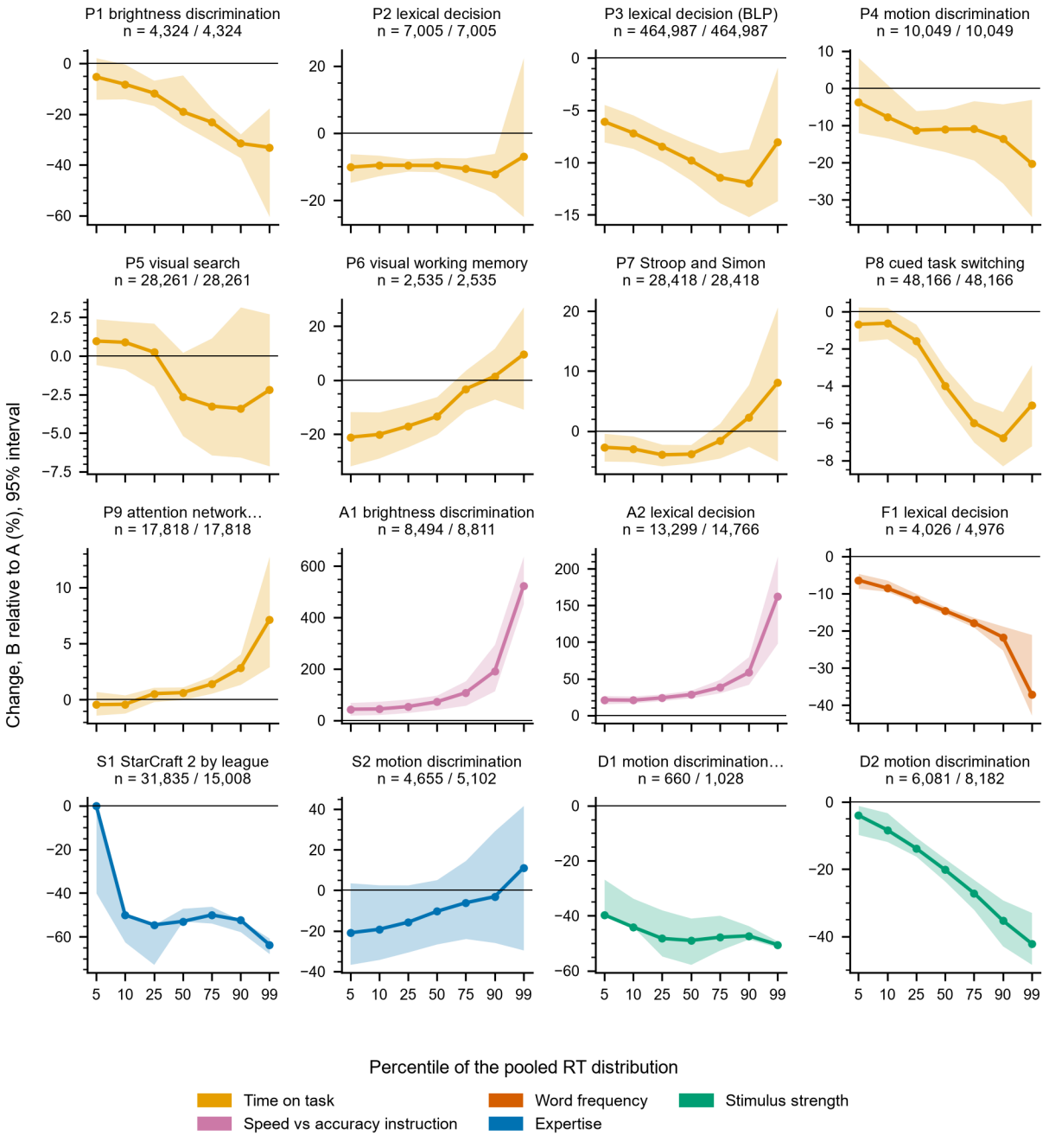


Figure 1. The data. Percent change from A to B at each percentile of the pooled RT distribution, every contrast, with 95% cluster-bootstrap intervals over units (bands); n = trials in A / B. Each panel has its own y scale.

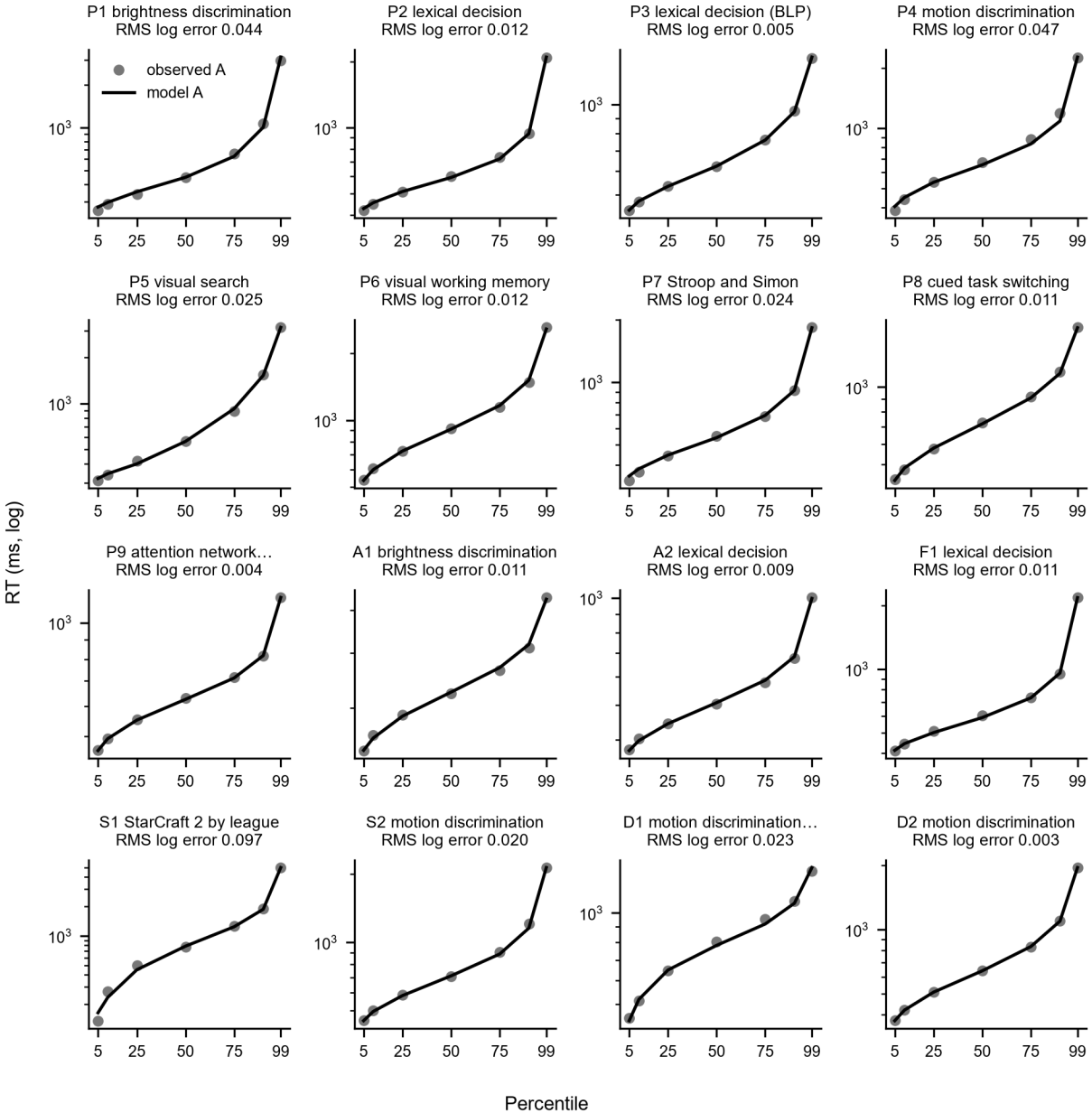


Figure 2. Step 1, the starting point. The model's condition A after calibration (line) against the observed pooled A (points). RMS log error 0.05 is about 5% at a typical percentile. Calibration moved four parameters from the WW-CarefulFitting fit of the nearest condition (2_data_processing/baselines.csv).

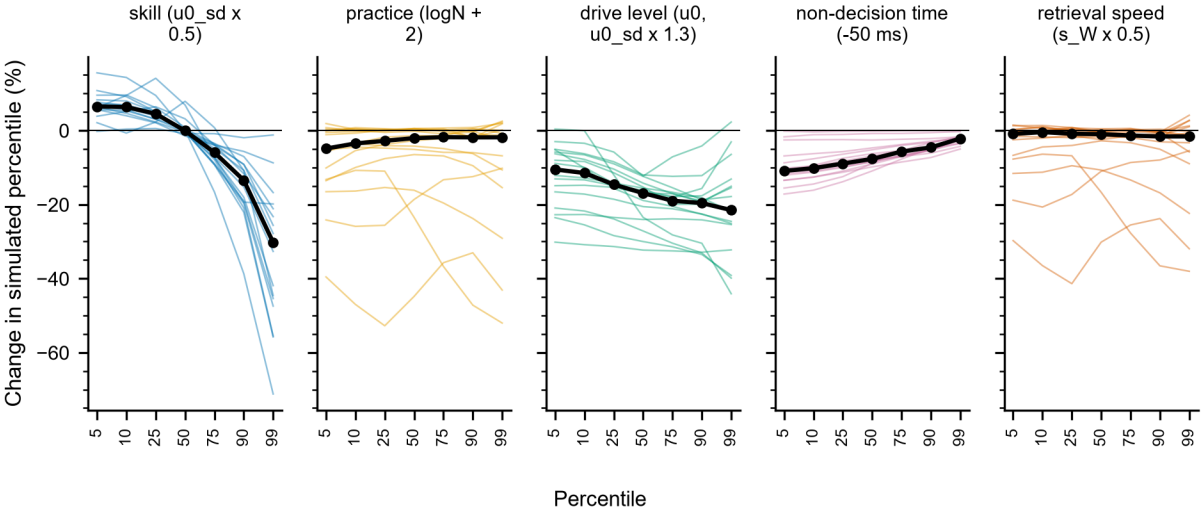


Figure 3. What each ingredient does. One step of each change applied to every contrast's calibrated A (thin lines; black = median over the 16 starting points). Ingredients that move the percentiles in the same pattern cannot be told apart by these data; the patterns differ most at the 5th-10th and 90th-99th percentiles.

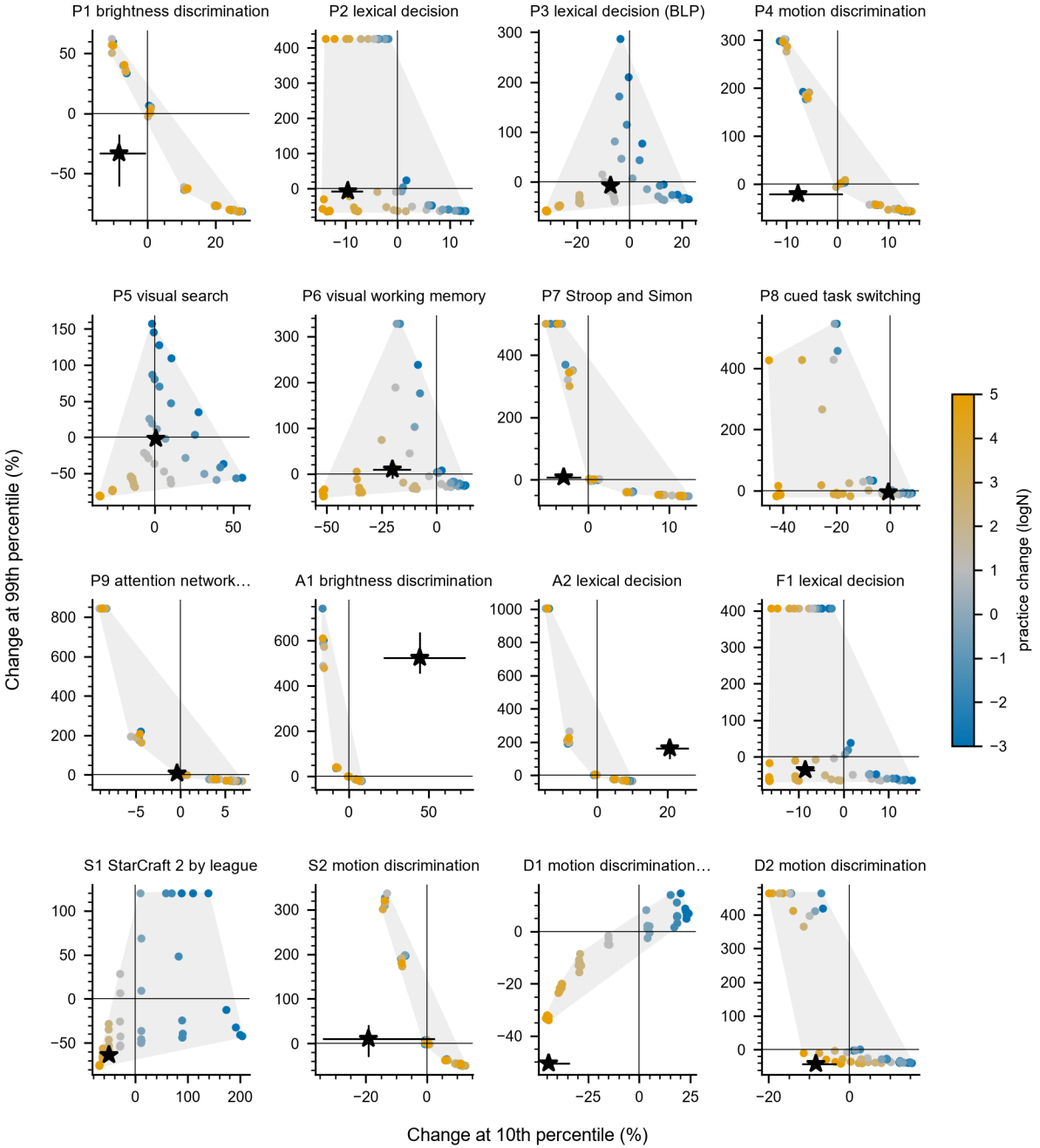


Figure 4. Step 3, what any mix can reach. Grey area: every change at the 10th and 99th percentiles that skill (u0_sd x 0.14 to x 2.7) and practice (logN -3 to +5) produce together from the calibrated A, on a 7 x 7 grid (points coloured by the practice change). Star: the observed change with its 95% interval. A star outside the grey area cannot be produced by these two mechanisms in this range. Points near +400% at the 99th percentile are mixes under which some simulated trials do not commit within 10 s.

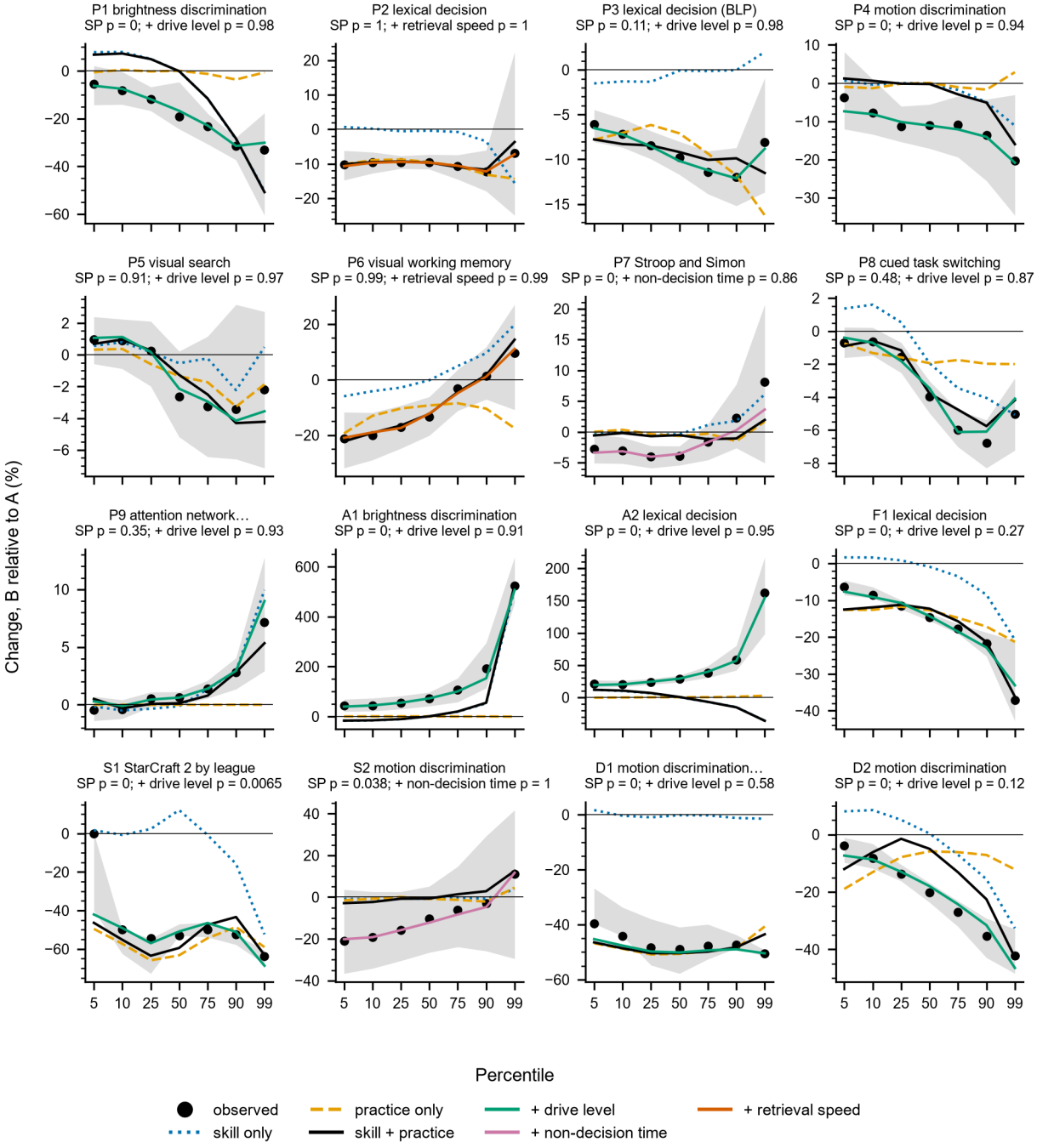


Figure 5. Step 2, the fits. Observed change (points, grey band = 95% interval) and the best fit of each model: skill only, practice only, both, and both plus the addition with the lowest AIC for that contrast (coloured as in the legend). p: chi-square adequacy; p > .05 = adequate.

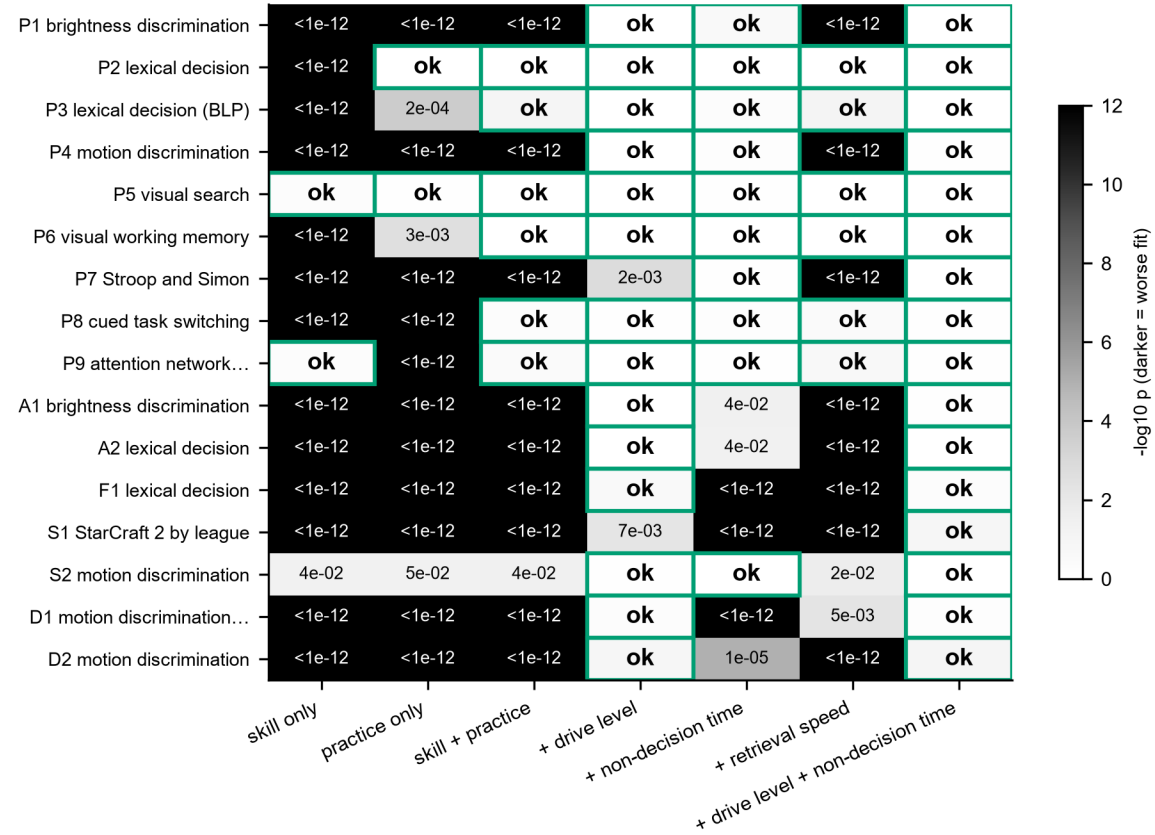


Figure 6. Which models are adequate. Each cell: chi-square p of a model for a contrast; ok in a green box = adequate ($p > .05$). Shading: -log10 p, capped at 12.

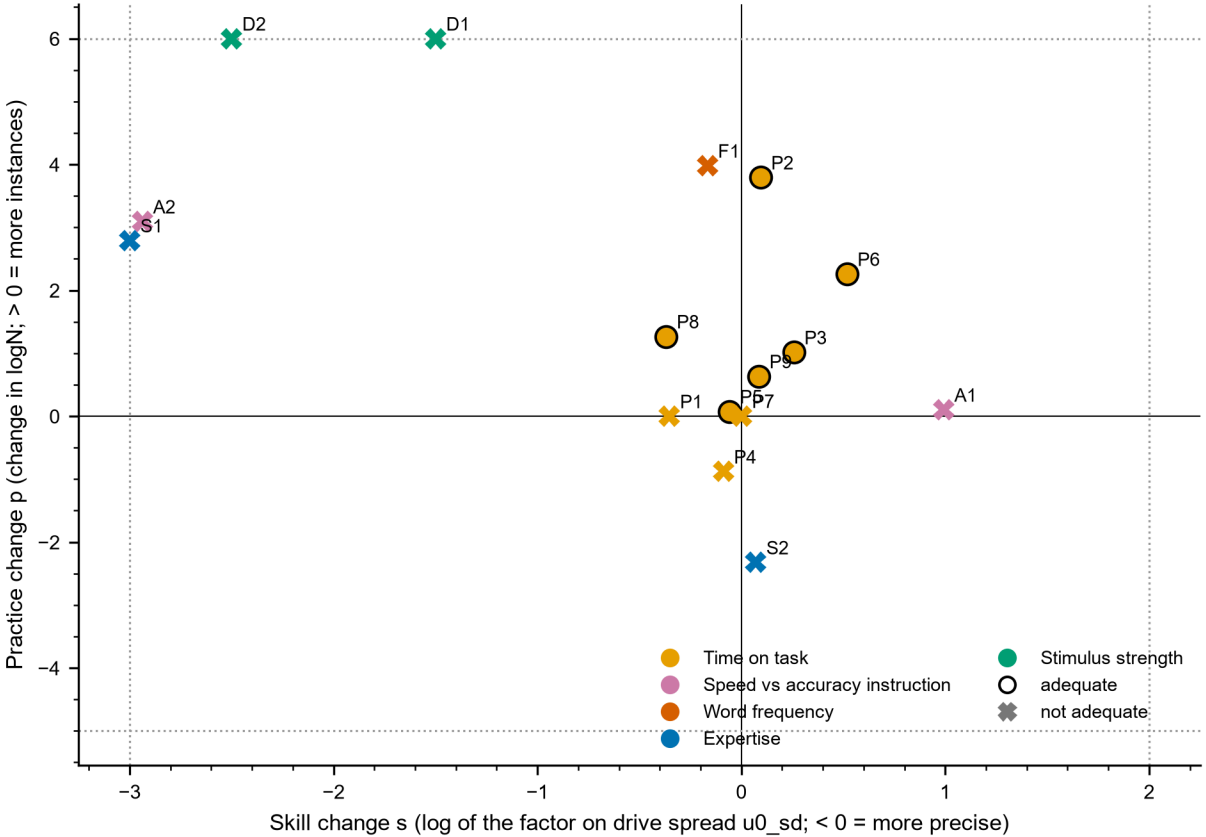


Figure 7. How much skill and how much practice the skill + practice fit uses for each contrast. Dotted lines: search bounds; a point on a bound used all the change allowed. Circle = adequate fit; cross = not adequate.

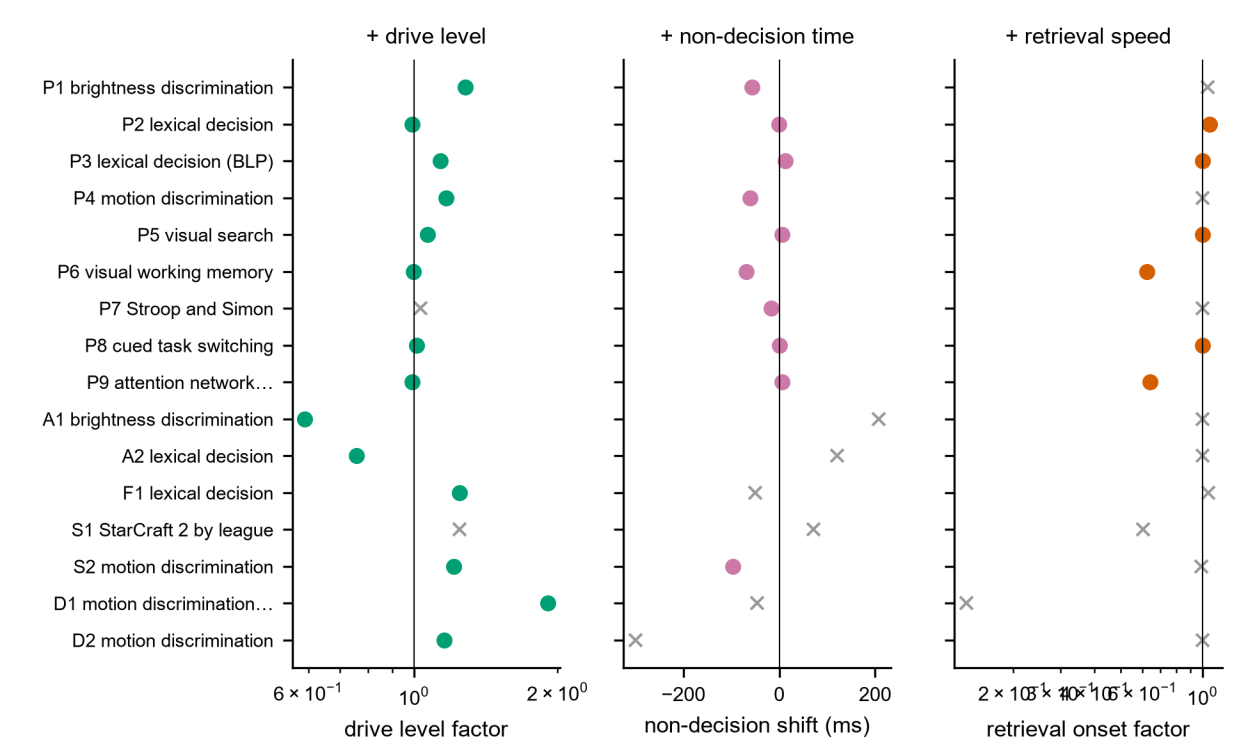


Figure 8. The added change in each three-change model: drive level (factor on u0 and u0_sd), non-decision time (ms), retrieval onset scale (factor; < 1 = faster). Filled circle = the model is adequate; grey cross = not.

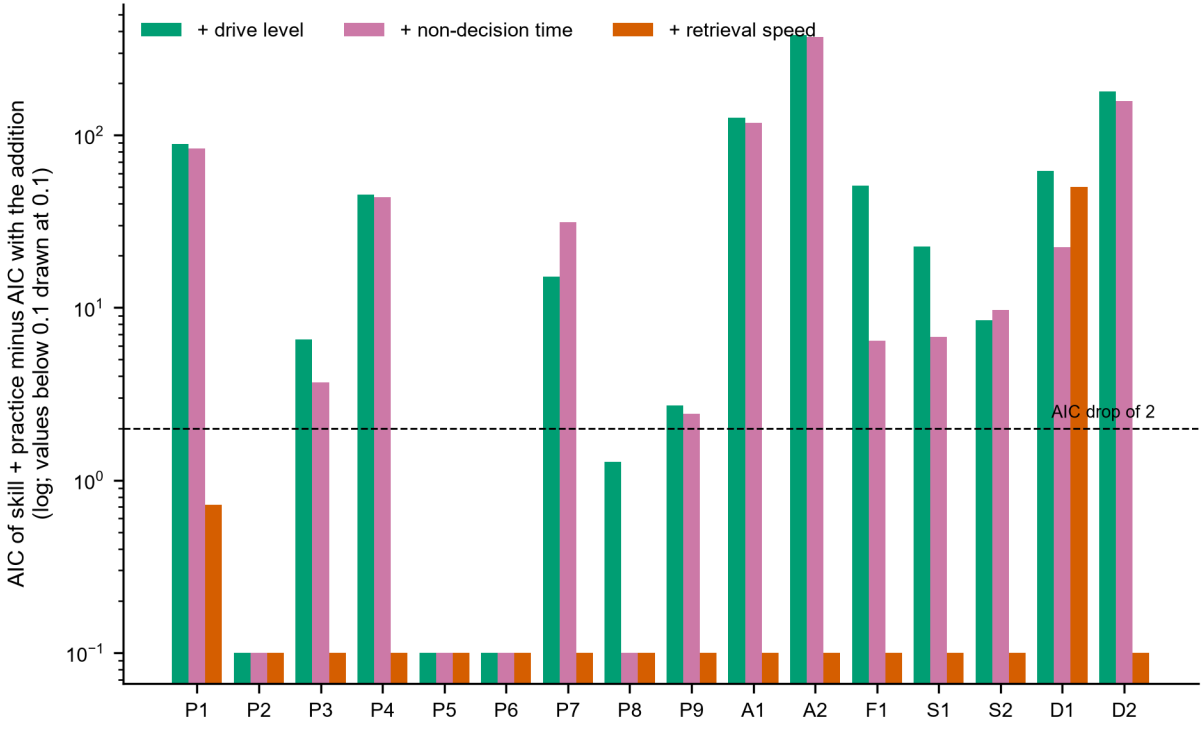
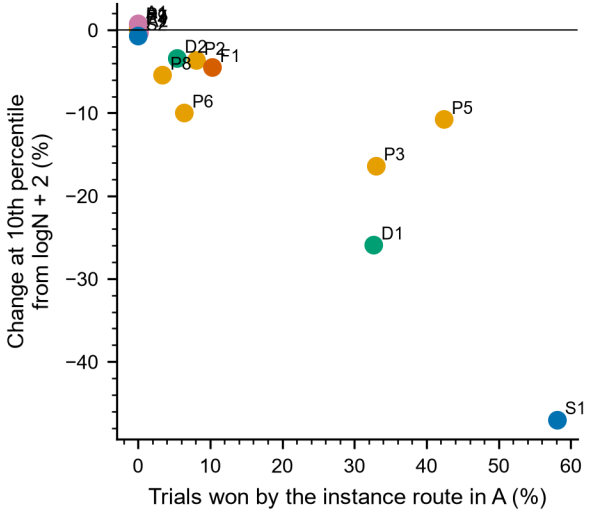


Figure 9. How much each addition improves on skill + practice, as the drop in AIC (larger = better fit for the extra parameter). Bars below the dashed line improve AIC by less than 2.



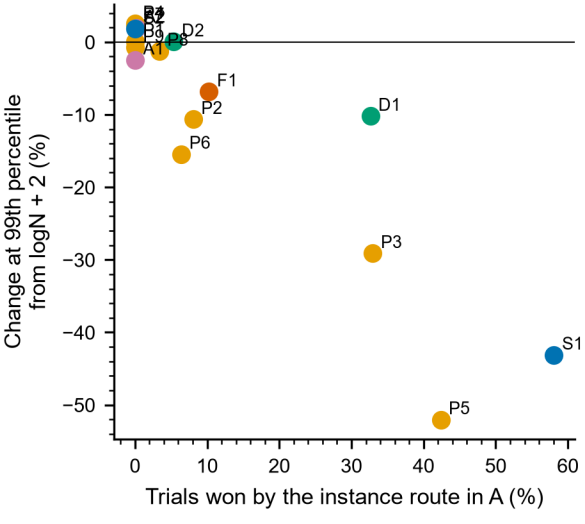


Figure 10. Practice works only through the instance route: the effect of one practice step (logN + 2) at the 10th and 99th percentiles against the share of A's simulated trials the instance route wins.